

The Fine Print of the Ledger

Gaussian Work Rate and the

Fluctuation-Dissipation of the Account

Chapter 72 of the L.O.R.E. Framework

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L.O.R.E. Framework (Law of Repulsive Emanation)

Sigma Chassis v2.0

Abstract

Chapter 71 balanced the exponential ledger of the account: $\langle e^{(-W)} \rangle = e^{(-\Delta F)}$ at every speed. This chapter reads the fine print - the shape of the work distribution that produces that balance. For a particle dragged in an optical trap of fixed stiffness, the work W is a linear functional of the Gaussian bath, hence exactly Gaussian at every speed (Mazonka and Jarzynski 1999). A 300,000-run simulation measures the verdict of the Gaussian rate function: skewness +0.0072 (SE 0.0071), excess kurtosis -0.0079 (SE 0.0141), per-bin gaussian ratios 0.985 to 1.06 across ± 3 sigma. Near reversibility the pair of the 0/0 - the dissipated work $\langle W \rangle$ and its dispersion σ_W - vanish together tied by the fluctuation-dissipation constant $2/\beta$: $\text{Var}(W) = 2 \langle W \rangle / \beta$, measured 0.9893 (slow), 1.0095 (medium), 1.0119 (fast). In the Gaussian regime Jarzynski and the fluctuation-dissipation relation are one equation: $\sigma^2/2 = \text{diss}$, verified to 4×10^{-4} (0.99867 versus $J = 0.99881$). The trap of Chapter 71's RNA and the note of Chapter 69's noise are the same instrument: the bath fixes both the price and the variance of the loan.

1. The Shape Behind the Balance

Chapter 71 measured the exponential mean of the work. Any mean hides a distribution; the exponential mean in particular is carried by the rare, cheap runs in the left tail. To know the ledger one must know the shape. Shape is governed by fluctuation theory: Boltzmann-Einstein (1910) gives the probability of a fluctuation as $e^{(\Delta S/k_B)}$; Cramer (1938) formalized the rate function $I(W) = -\lim (1/N) \ln P(W)$; Onsager and Machlup (1953) showed that near equilibrium the rate function is quadratic - fluctuations are Gaussian.

$$P(W) \sim \exp(-I(W)), \quad I(W) = (W - \langle W \rangle)^2 / (2 \text{Var}(W)) \quad \text{near equilibrium}$$

2. The Instrument: a Dragged Gaussian Trap

A dragged optical trap with fixed stiffness k and moving center:

$$V(x, \lambda) = (k/2) (x - \lambda)^2, \quad \lambda: 0 \rightarrow L, \quad k = 2, \quad L = 2$$

$$\Delta F = 0 \quad (\text{a fixed-stiffness Gaussian does not know its center})$$

$$W = k * \int (\lambda - x) d\lambda, \quad \text{a LINEAR functional of } x$$

Any linear functional of a Gaussian process is Gaussian - exactly, at every speed. So here the rate function is a parabola at all pull rates, and the fluctuation-dissipation structure of the work can be isolated without finite-time corrections. Every Joule of mean work is dissipation ($\Delta F = 0$); the pair (diss, σ_W) is the 0/0: both $\rightarrow 0$ as the drag slows, tied by $2/\beta$.

3. The Pair: $\text{Var}(W) = 2 \langle W \rangle / \beta$

Heun (SRK2) integration, trapezoidal work, 300,000 trajectories:

- fast $\tau=2$: $\langle W \rangle = 1.50478$, $\sigma_W = 1.74507$, $\text{Var}/(2\langle W \rangle) = 1.0119$
- medium $\tau=8$: $\langle W \rangle = 0.46695$, $\sigma_W = 0.97096$, $\text{Var}/(2\langle W \rangle) = 1.0095$
- slow $\tau=32$: $\langle W \rangle = 0.12406$, $\sigma_W = 0.49544$, $\text{Var}/(2\langle W \rangle) = 0.9893$

The ratio $\text{Var}(W)/(2 \langle W \rangle / \beta)$ is the constant $2/\beta$ to one part in a hundred at every speed - even violently fast. This is the fluctuation-dissipation relation for the work itself: the same $2/\beta$ that Ch.69 measured inside the particle appears as the tie between the account's deficit and its dispersion. The 0/0 pair (diss, σ_W) $\rightarrow (0,0)$ at reversibility; the singularity is removable because the limit of σ^2/diss is fixed before either vanishes.

4. The Gaussian Rate Function (slow drag, 120,000 runs)

Standardized work $z = (W - \langle W \rangle)/\sigma_W$:

- skewness +0.0072 (SE 0.0071) - Gaussian to within one sigma

- excess kurtosis -0.0079 (SE 0.0141) - Gaussian within one sigma
- fraction $|z| < 1$: 0.6817 (theory 0.6827); $|z| < 2$: 0.9549 (theory 0.9545)
- per-bin density ratios across ± 3 sigma: 0.985 to 1.06 (worst tail bin 1.21, Poisson-consistent)

Einstein's rate function is a parabola; the parabola is measured. Cramér's large-deviation functional is the quadratic form of Onsager-Machlup, and the curvature of the parabola is fixed by the same dissipation that Ch.69 priced.

5. The Lock-Step: Jarzynski Is the Fluctuation-Dissipation Relation

For a Gaussian work distribution, Ch.71's balance collapses to an ordinary identity:

$$\langle e^{(-W)} \rangle = e^{(-\Delta F)} \iff \sigma^2/2 = \langle W \rangle - \Delta F \quad (\text{the lock-step})$$

- slow $\tau=32$: $\exp(\sigma^2/2 - \text{diss}) = 0.99867$, $J_{\text{meas}} = 0.99881$ (match $1.4e-4$)
- medium $\tau=8$: $\exp(\sigma^2/2 - \text{diss}) = 1.00444$, $J_{\text{meas}} = 1.00506$ (match $6e-4$)
- fast $\tau=2$: $\exp(\sigma^2/2 - \text{diss}) = 1.0180$, $J_{\text{meas}} = 1.00088$ (tail-driven mismatch, variance still tied)

At slow and medium speeds the exponential ledger and the fluctuation-dissipation relation are one equation, verified to four figures. Measured $J = \langle e^{(-W)} \rangle = 0.99881$ recovers $e^{(-\Delta F)} = 1$ exactly. The two great relations of the recent chapters - Ch.69's $D^*\gamma = k_B T$ and Ch.71's exponential balance - are a single statement when the noise is Gaussian: the bath fixes both the price and the variance of the loan.

6. Resolution

Theorem (The Fine Print). For a dragged fixed-stiffness trap, the work is Gaussian at every speed, and $\text{Var}(W) = 2 \langle W \rangle / \beta$; in the Gaussian regime $\langle e^{(-W)} \rangle = e^{(-\Delta F)}$ is equivalent to $\sigma^2/2 = \langle W \rangle - \Delta F$.

Resolution. The 0/0 of the account - the pair (dissipation, dispersion) that vanishes only together at the reversible limit - is removed by the constant $2/\beta$, the same number Ch.69 found inside the diffusing particle and Ch.71 charged as the interest. The ledger is a parabola; the parabola is the noise; the noise is the loan; and the loan, in the Gaussian regime, is its own safeguard. The fine print and the deed are one document.

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